EduBridge

Connecting Retired Teachers with Students

Complete Project Report & Technical Blueprint

Prepared for: Academic Submission

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1. Executive Summary

EduBridge is a web-based platform designed to bridge the gap between retired, highly experienced schoolteachers and students who need quality, personalized tutoring. The core idea is simple: thousands of retired teachers are sitting at home with decades of pedagogical knowledge and nothing meaningful to do with it — while students across the country struggle to find affordable, experienced educators.

This platform enables students to search for retired teachers by subject, language, availability, and rating; send them messages; book one-on-one or group sessions; and conduct live video calls — all within a single, secure, easy-to-use interface. Teachers, in turn, get a structured way to stay engaged, contribute to society, and optionally earn supplemental income.

Aspect	Detail
Platform Name	EduBridge
Core Problem	Students lack experienced tutors; retired teachers lack purpose & platform
Primary Users	Students (K–12, college), Retired Teachers, Admins
Key Features	Profiles, Search & Filter, Real-Time Chat, Video Calls, Booking, Payments
Tech Stack	Laravel (PHP), Vue.js, MySQL, Pusher, Twilio / WebRTC
Deployment	Cloud (AWS / DigitalOcean), Docker, HTTPS
Revenue Model	Freemium + Commission on paid sessions + Premium subscriptions

2. Problem Statement

2.1 The Student Side

Students — especially those from small towns, rural areas, or lower-income families — often cannot access quality tutoring. Private tutors are expensive, and school teachers are overloaded. Many students specifically want subject matter experts with years of classroom experience, not just a peer tutor or an online video.

2.2 The Retired Teacher Side

India alone has millions of retired government and private school teachers. After retirement, many feel socially isolated, purposeless, and underutilized. They have immense subject knowledge and teaching skills but no structured channel to apply them. They are not comfortable finding gigs on generic platforms like Upwork.

2.3 The Missing Bridge

There is no dedicated, trustworthy platform that:

- Verifies retired teacher credentials and backgrounds
- Allows students to browse teacher profiles and reach out
- Facilitates real-time messaging and video sessions securely
- Manages bookings, reminders, and feedback — all in one place

EduBridge fills this gap completely.

3. Target Users & Personas

3.1 Students

- Age group: 10 to 25 years (school students, college undergraduates)
- Need: Affordable, experienced tutoring in specific subjects
- Pain point: Cannot find trusted retired experts; existing platforms have young tutors only
- Device usage: Smartphone (primary), laptop/tablet

3.2 Retired Teachers

- Age group: 55 to 75 years
- Background: Retired government/private school or college professors
- Need: Meaningful engagement, social connection, optional income
- Pain point: Not tech-savvy; need a simple, guided interface
- Motivation: Give back to society, stay mentally active

3.3 Admins / Platform Operators

- Verify teacher credentials and approve profiles
- Moderate reported content and disputes
- Monitor platform analytics and usage metrics

4. Complete Feature List

4.1 User Registration & Authentication

- Separate registration flows for Students and Teachers
- Email/phone OTP verification for account activation
- Social login (Google) for students
- Password reset via email link
- Role-based access control (Student / Teacher / Admin)
- Profile completeness indicator and guided onboarding

4.2 Teacher Profile System

- Full name, profile photo, short bio
- Educational qualifications (degree, institution, year)
- Teaching subjects (multi-select: Math, Science, English, History, etc.)
- Years of experience and previous school/college affiliation
- Languages they can teach in (Hindi, Punjabi, English, etc.)
- Hourly rate (or 'Free / Volunteer' option)
- Availability calendar — days and time slots they are free
- Star ratings and student reviews visible on profile
- Badge system: 'Verified', 'Top Rated', 'New Teacher'

4.3 Student Profile System

- Name, class/grade, school name
- Subjects they need help in
- Preferred language of instruction
- Saved/bookmarked teacher profiles
- Session history and upcoming bookings
- Payment method on file

4.4 Smart Search & Matching

- Search bar: filter by subject name, teacher name
- Advanced filters: subject, language, price range, rating, availability (day/time), gender preference
- AI-assisted recommendations: 'Teachers you might like based on your searches'
- Sort by: Relevance, Rating (High to Low), Price (Low to High), Experience
- Paginated results with teacher cards showing key info at a glance

4.5 Real-Time Messaging

- Students can send the first message to any teacher (like an inquiry)

- Teachers can accept or decline conversations
- WebSocket-based chat — messages delivered instantly (no page refresh needed)
- Support for text messages, image/file attachments (e.g., homework photos, PDFs)
- Message read receipts (single tick = sent, double tick = seen)
- Chat history stored and searchable
- Typing indicators ('Teacher is typing...')
- Message notifications via browser push notification and email digest

4.6 Group Messaging

- Teacher can create a 'Class Group' and add multiple students
- Group chat for announcements, sharing notes, homework reminders
- Students can ask questions in the group; teacher replies publicly
- Admin controls for the group: mute, remove student, delete messages

4.7 Video Calling (One-on-One & Group)

- 'Start Session' button in the booking or chat screen launches a video room
- One-on-one sessions: teacher and single student
- Group sessions: teacher + up to 20–30 students simultaneously
- Features: Camera on/off, microphone mute/unmute, screen sharing
- In-session text chat sidebar
- Session recording (optional, with consent from both parties)
- Raise hand feature for group sessions (student signals they want to speak)
- Whiteboard / collaborative drawing tool for math/diagram explanations
- Session timer visible to both parties

4.8 Booking & Scheduling System

- Teacher marks available time slots on an integrated calendar
- Student browses teacher's available slots and books a session
- Booking confirmation sent to both parties via email and SMS
- 24-hour reminder notification before session
- Cancellation policy: student can cancel up to 2 hours before; teacher can cancel up to 1 hour before
- Rescheduling option with mutual consent
- Session history: completed, upcoming, cancelled

4.9 Ratings & Reviews

- After each session, student rates teacher (1–5 stars) and leaves a written review
- Teacher can optionally rate/comment on the student's engagement
- Reviews are publicly visible on teacher profile
- Admin can flag and remove fake or abusive reviews

4.10 Notifications System

- In-app notification bell (new messages, bookings, reminders)
- Email notifications (booking confirmations, session reminders, new reviews)
- SMS/WhatsApp notifications for older teachers who prefer it
- Notification preferences: user can choose which alerts to receive

4.11 Payment System (for paid sessions)

- Students prepay for sessions via Razorpay / Stripe / UPI
- Funds held in escrow; released to teacher after session completion
- Platform commission: 10–15% per paid session
- Teacher dashboard: earnings overview, payout history
- Automatic payout to teacher's bank account weekly
- Refund policy: full refund if teacher cancels; partial if student cancels late
- Free/volunteer sessions are always available (no payment required)

4.12 Admin Panel

- User management: view, edit, suspend, or delete any user account
- Teacher verification queue: review submitted credentials and approve/reject
- Content moderation: flag, review, and remove reported messages or reviews
- Analytics dashboard: DAU, MAU, sessions conducted, revenue, top teachers
- Announcement system: push platform-wide notifications
- Dispute resolution: mediate between student and teacher on refund/cancellation issues

5. Technology Stack

Layer	Technology	Purpose
Backend Framework	Laravel 10 (PHP 8.2)	MVC structure, routing, auth, Eloquent ORM
Frontend	Vue.js 3 + Blade templates	Reactive UI components, SPA-like experience
Real-Time Chat	Laravel Echo + Pusher	WebSocket broadcasting for instant messaging
Video Calls	Twilio Programmable Video / Daily.co	Scalable video sessions, recording, screen share
Database	MySQL 8 (primary), Redis (cache/sessions)	Data storage and fast caching
Authentication	Laravel Breeze + Spatie Permissions	Login/register, OTP, role-based access
Scheduling	Laravel Zap / custom calendar engine	Teacher availability and booking management
Payments	Razorpay (India) / Stripe (international)	Session payments, escrow, payouts
Notifications	Laravel Queues + Mailgun + Firebase FCM	Email, push, SMS notifications
File Storage	AWS S3 / Cloudflare R2	Profile photos, attachments, session recordings
Deployment	Docker + AWS EC2 / DigitalOcean	Containerized, scalable cloud hosting
CDN	Cloudflare	Static assets, DDoS protection, SSL
Search	Laravel Scout + MeiliSearch	Fast full-text teacher search and filters
Testing	PHPUnit + Laravel Dusk	Unit, feature, and browser automation tests

6. Database Design

The relational database (MySQL) is the backbone of all user data, sessions, and transactions. Below are the primary tables and their key columns.

6.1 Core Tables

Table	Key Columns	Notes
users	id, name, email, phone, password, role (student/teacher/admin), avatar, status, created_at	Central auth table; role determines dashboard
teacher_profiles	id, user_id (FK), bio, subjects (JSON), experience_years, languages (JSON), hourly_rate, is_verified, rating_avg, total_reviews	Extended profile for teachers only
student_profiles	id, user_id (FK), class_grade, school_name, subjects_needed (JSON), preferred_language	Extended profile for students only
teacher_availability	id, teacher_id (FK), day_of_week, start_time, end_time, is_recurring, specific_date	Defines when teacher is available
bookings	id, student_id, teacher_id, start_at, end_at, status (pending/confirmed/completed/cancelled), session_type (solo/group), price, payment_status	Core booking record
conversations	id, created_by, title (for groups), is_group, created_at	Chat thread — 1:1 or group
conversation_participants	id, conversation_id, user_id, joined_at, left_at	Who is in each conversation
messages	id, conversation_id, sender_id, body, type (text/image/file), read_at, created_at	Individual messages
video_sessions	id, booking_id, room_name, twilio_room_sid, started_at, ended_at, recording_url	Video call metadata
reviews	id, booking_id, reviewer_id, reviewee_id, rating (1-5), comment, created_at	Post-session ratings
payments	id, booking_id, payer_id, amount, platform_fee, teacher_payout, gateway, gateway_ref, status, paid_at	Payment ledger

Table	Key Columns	Notes
notifications	id, user_id, type, data (JSON), read_at, created_at	In-app notification store

7. System Architecture

7.1 High-Level Architecture

EduBridge follows a monolithic Laravel MVC architecture for the initial version, with clear separation of concerns. The frontend (Vue.js components embedded in Blade views) communicates with the Laravel backend via RESTful API routes and broadcasts real-time events via Pusher.

Layer	Components
Client Layer	Web Browser (Vue.js + Blade), Mobile Browser (PWA), Future: iOS/Android App
API Gateway	Laravel Routes (web.php, api.php) — REST endpoints + WebSocket event broadcasting
Application Layer	Controllers, Services, Jobs (Queue Workers), Events & Listeners
Data Layer	MySQL (persistent data), Redis (sessions, cache, queue), AWS S3 (files)
External Services	Pusher (WebSocket), Twilio (Video), Razorpay (Payments), Mailgun (Email), Firebase (Push)

7.2 Key Architectural Flows

Chat Message Flow

1. Student types message → HTTP POST to /api/messages
2. MessageController validates, saves to DB, fires MessageSent event
3. Event broadcast via Pusher to all channel subscribers
4. Teacher's browser (Vue.js) listens via Laravel Echo → message appears instantly
5. Push notification dispatched to queue worker → email/SMS/FCM sent

Video Session Flow

6. Student clicks 'Join Session' on a confirmed booking
7. Backend calls Twilio API to create a video room, generates access token
8. Token returned to Vue.js frontend, which connects to Twilio video room
9. Both participants see each other; session timer starts
10. On session end, backend records completion, triggers payment release

8. Detailed User Flows

8.1 Student Journey

- Sign up → Verify email/phone → Complete student profile
- Search for teacher by subject (e.g., 'Class 12 Math') → Apply filters (Hindi medium, free, available evenings)
- View teacher profile: qualifications, reviews, price, availability
- Send an intro message: 'Namaste sir, mujhe integration ka concept samajhna hai'
- Teacher responds → Conversation established
- Student books a 1-hour session slot from teacher's available calendar
- Razorpay payment (or free session) → Booking confirmed
- Email + SMS reminder 24 hours before session
- Click 'Join Session' at scheduled time → Video call starts
- After session: rate teacher + write review

8.2 Teacher Journey

- Sign up → Submit credentials (degree certificates, service record)
- Admin verifies credentials → Profile gets 'Verified' badge
- Complete profile: bio, subjects, languages, availability slots, hourly rate (or 'Free')
- Receive message notification from interested student
- Reply and agree to book a session
- Accept booking request → Calendar slot blocked
- Join video session at scheduled time
- Post-session: earnings reflected in dashboard (for paid sessions)
- View and respond to student reviews

9. Unique Value Proposition & Impact

9.1 What Makes EduBridge Different

Feature	EduBridge	Generic Tutoring Apps
Tutor type	Verified, retired professional teachers	Anyone can register as a tutor
Credential verification	Mandatory admin review of degrees & service records	Usually self-declared
Free/volunteer option	Always present — core mission	Rare; mostly paid only
Language support	Regional Indian languages (Hindi, Punjabi, etc.)	Mostly English-only
User interface for elders	Simple, large text, guided onboarding for 55+ users	Not optimized for seniors
Group sessions	Built-in — one teacher to many students	Often extra cost or missing
Intergenerational focus	Core mission and design principle	Not a focus
Community features	Forums, study groups, shared notes	Transactional only

9.2 Social Impact

- Students in rural areas and small towns get access to city-level teaching expertise
- Economically weaker students get free tutoring from volunteer retired teachers
- Retired teachers experience renewed purpose, mental engagement, and social connection
- Reduces the social isolation problem common among senior citizens
- Preserves decades of pedagogical knowledge that would otherwise be lost
- Scalable to lakhs of users — impact multiplies as platform grows

10. Business Model & Monetization

10.1 Revenue Streams

Revenue Stream	Description	Estimated Revenue
Commission on Paid Sessions	10–15% platform fee on each paid session booking	High volume, core revenue
Teacher Premium Subscription	₹299/month for featured profile listing, analytics, priority search placement	Recurring, predictable
Student Pro Plan	₹99/month: unlimited bookings, group session access, recording downloads	Recurring
Institutional Partnerships	Schools/colleges pay for bulk session access for their students	B2B, high value
Advertising (non-intrusive)	Educational brands (books, stationery, courses) can advertise	Secondary source

10.2 Free Tier (Always Available)

The platform will always maintain a free tier where volunteer retired teachers can offer their time at no cost. This is central to the social mission. Payment features are opt-in for teachers who want compensation.

11. Development Roadmap

Phase	Timeline	Deliverables
Phase 1 — Foundation	Months 1–2	Laravel setup, DB schema, user registration, role-based auth, teacher/student profiles
Phase 2 — Core Features	Months 3–4	Teacher search & filters, real-time chat (Pusher), booking & calendar system
Phase 3 — Video & Payments	Months 5–6	Twilio video integration, Razorpay payments, escrow system, email/SMS notifications
Phase 4 — Group Features	Month 7	Group messaging, group video sessions, whiteboard tool, session recordings
Phase 5 — Admin & Moderation	Month 8	Admin panel, credential verification flow, review moderation, analytics dashboard
Phase 6 — Polish & Beta	Month 9	UI/UX refinements, PWA, performance optimization, security audit, beta testing
Phase 7 — Launch & Scale	Month 10+	Public launch, marketing, CDN, horizontal scaling, mobile app planning

12. Security & Privacy Considerations

- HTTPS enforced across all routes via SSL/TLS (Cloudflare)
- All passwords hashed using bcrypt (Laravel default)
- SQL injection prevention via Eloquent parameterized queries
- CSRF protection on all form submissions (Laravel CSRF tokens)
- Input validation and sanitization on all user inputs
- Rate limiting on login, registration, and message-sending endpoints
- Private chat channels authenticated via Laravel Echo presence channels
- Video rooms accessible only via time-limited Twilio access tokens
- Student age verification — platform is designed for minors, so teacher background checks are required
- GDPR/PDPB (India) compliance: users can request data deletion
- Session recordings stored with AES-256 encryption; accessible only to participants
- Admin two-factor authentication mandatory

13. Challenges & Mitigation Strategies

Challenge	Risk Level	Mitigation Strategy
Retired teachers not tech-savvy	High	Simplified UI, large fonts, onboarding videos in Hindi/regional languages, WhatsApp helpline
Trust & safety (minors on platform)	High	Mandatory teacher credential verification, report button, admin moderation, no anonymous users
Low initial teacher supply	High	Partner with retired teachers' associations, government pension departments, word-of-mouth campaigns
Video call quality in rural areas	Medium	Adaptive bitrate streaming, fallback to audio-only mode on low bandwidth
Payment disputes	Medium	Escrow system, clear refund policy, admin dispute resolution, session completion verification
Fake teacher profiles	Medium	Document upload verification, AI fraud detection, community reporting
Scaling to large user base	Medium	Horizontal scaling, CDN, DB read replicas, queue workers, caching (Redis)
Monetization vs social mission tension	Low	Free volunteer tier always maintained; paid features are add-ons, not gates

14. Future Enhancements

- Native iOS and Android mobile apps (React Native)
- AI-powered tutor matching: algorithm suggests best teacher based on student's learning history and performance
- Multilingual platform UI (Hindi, Punjabi, Tamil, Telugu, Bengali, Marathi)
- Gamification: badges for teaching hours, student streaks, leaderboard
- Study material sharing: teachers can upload notes/PDFs for enrolled students
- Online tests and quizzes within the platform
- Parent dashboard: parents of minor students can monitor session activity and progress
- Integration with government school databases for direct teacher verification
- Offline lesson downloads for areas with intermittent internet
- Analytics for teachers: see which explanations worked, student engagement score

15. Conclusion

EduBridge is more than a tutoring platform — it is a social infrastructure project. It unlocks the immense, currently-idle knowledge of retired educators and channels it towards students who need it most. By combining a robust technical foundation (Laravel, Vue.js, Twilio, Pusher) with a thoughtful user experience designed for both young students and elderly teachers, EduBridge creates a sustainable, scalable, and socially impactful ecosystem.

The platform addresses a real, urgent, and massive problem in the Indian education landscape — and has strong potential for regional and international expansion. It generates revenue while maintaining a social mission, making it both commercially viable and deeply purposeful.

"Retired teachers have taught for decades. EduBridge gives them the stage to teach for decades more."